Datasheet

Black Oak Engineering

precision electronic instruments

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Monarch Accelerometer

3-axis ±200g

Part number BE-MAC1
Version 1.1

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Description. The Monarch-Accel-200g (#BE-MAC1) is a simple, rugged, compact digital accelerometer that measures up to ±200 g in three axes simultaneously, while outputting a data stream via USB. It is small enough to fit into tight spaces, and has a rigid mechanical design with mounting flanges. The #BE-MAC1 is typically used to measure moderate-level shock and vibration events. Filtering is easily programmable. The #BE-MAC1 connects via a USB-C cable (included). This provides power and communications.

The Monarch family is simple to use. Just connect the USB-C cable. Use any serial port terminal emulation software that you prefer (PuTTY, Tera Term, etc.) See User Guide for the simple instruction set. For PC, Mac, Linux, Raspbian, etc. No complicated, messy app installs. No subscriptions. No 'ecosystem'. Just reliable data.

Basic specifications

- Range: −200 to +200 g (1 g = 9.81 m/s^2) in 3 axes.
- Axes indicated on enclosure.
- Precision: 100 mg.
- RMS Noise: 3 LSB each axis.
- Sample rate: 400 to 6400 Hz programmable.
- High pass filter range: 0.24 to 30.48 Hz.
- Low pass filter range: 200 to SR / 2 Hz, 4-pole.
- Shielded low-noise construction.
- Mass 14.9 grams.
- Current: 5mA normal operation.
- Standard 1 m USB-C cable.
- Communication is serial stream via USB. For greater programmability you may use the Acorn API. This is an easy extensible data layer.
- Temperature range: −30 to +80 °C (−22 to +175 °F).
- Compact size: 35 L × 35 W × 15 H mm. (1.4 L × 1.4 W × 0.6 H inch), with mounting flanges.
- EAN-13 0729043992836

Monarch butterfly (***Danaus plexippus***) They migrate annually from Mexico and the southern US to the northern areas, often 1800 miles (2900 km) away. They lay their eggs in milkweed plants, which the larvae require. Their navigational skill depends primarily upon solar tracking and an accurate clock, but there is strong evidence that the monarch can also sense Earth's magnetic field.

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BOE is here to support you. If you need a modification, contact us. Our Engineering team can often make adjustments to the standard design.

BOE is continuously improving. We also strive to keep one step ahead of procurement shortfalls. We will deliver to you the latest hardware version possible. In some cases specifications will change.

Approvals & Compliance

- RoHS. REACH.
- California Prop 65. There are no listed toxic materials in the PCBA.
- CE. The system is designed and tested to meet or exceed CE requirements for laboratory, non-safety-critical electronic devices.

Warranty Policy. Any instrument ordered from BOE may be returned for full refund, less shipping costs, within 30 days of delivery, provided that the instrument has not, in the opinion of BOE been damaged or misused. An RMA number is required in all cases. See our *Standard Terms & Conditions – Instruments* for more details.

BOE reserves the right to make changes to these specifications as it deems necessary. All technical information contained herein is as accurate as possible; however BOE shall not be held responsible for any errors or for product use, nor for any infringements upon the rights of others which may result from its use. BOE products are not to be used in life support or safety critical applications.

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